

MA341 APPLIED ODE's

Lecture Notes

2 Direction Field

An equation of the type $y' = f$ describes a slope of the solution $y = y(t)$. This can be used to graphically study solutions behavior. The graphical representation reminds the graph of Vector Field from Calc 3 course, although those things are very different. Every trajectory on a direction field represents a solution to differential equation. Direction fields become very useful when there no explicit solution do a differential equation is available.

Example 2.1. Consider an equation

$$y' = x - y$$

. Clearly, when $x = y$ the slope is $y' = 0$, see Figure 1a blue arrows. If we fix x and take various values of y the slope become negative at some moment, and it's getting steeper and steeper, see Figure 1a black arrows. Similarly, fixing y and taking various values of x , the slope will turn positive and will get steeper as well, see Figure 1a red arrows. In the end one arrive at Figure 1b.

2.1 The method of isoclines - level curves

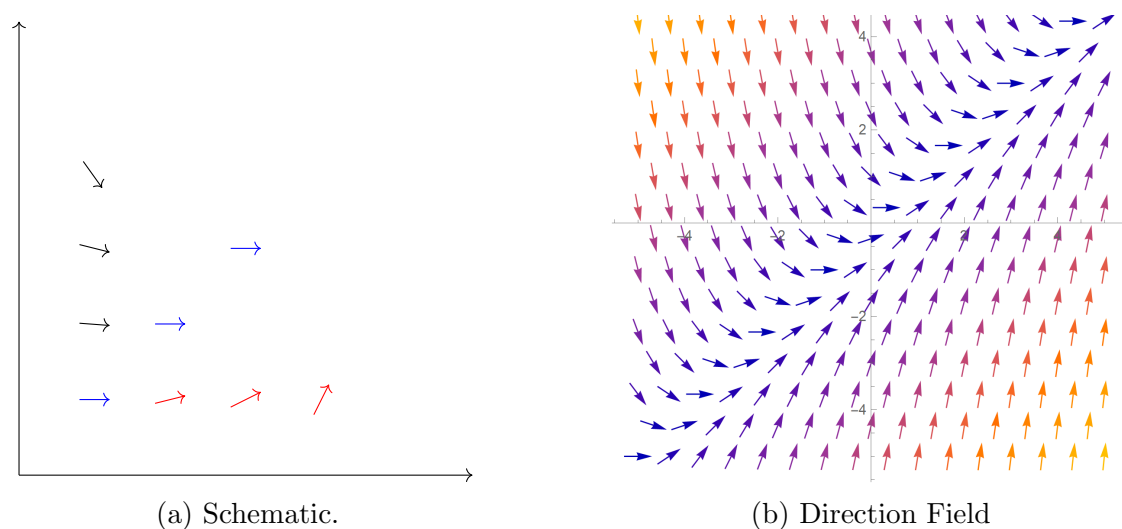
Consider an ODE

$$\frac{dy}{dx} = f(x, y).$$

Using different values of C draw a level curve $f(x, y) = C$. We usually use values of C that are easier to calculate/represent.

Example 2.2. Consider

$$\frac{dy}{dx} = xy.$$

Figure 1: Direction Field: $y' = x - y$.

Choosing the value of C will give us a level curve $xy = C$. Solve it for y to get a hyperbola

$$y = \frac{C}{x}.$$

Thus, on each hyperbola $y = \frac{C}{x}$ we have a slope C , see Figure 2a. Figure 2b shows a solution in case of $y(0) = 1$.

2.2 Phase Lines

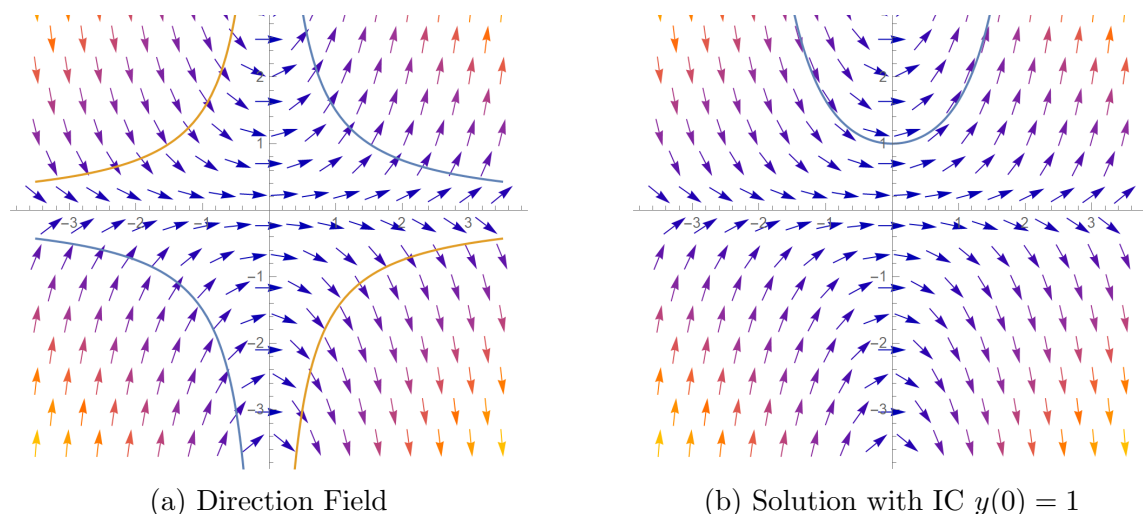
Definition 2.3. An autonomous (or time-invariant) differential equation is an equation that does not explicitly depend on the independent variable, i. e. the case of

$$\frac{dy}{dt} = f(y).$$

Definition 2.4. A solution to a differential equation is called an equilibrium solution if it is constant and the slope of an equation is horizontal, i. e. if $y = C$ ($C = \text{const}$) and $\frac{dy}{dt} = 0$, those are the roots of $f(C) = 0$.

We distinct 3 types of equilibria

1. A **sink** is a **stable** equilibrium, which is also its other name. Its attracting all

Figure 2: Direction Field: $y' = xy$.

neighboring solutions to it as $t \rightarrow \infty$. If y is a sink, then the solution $y + \epsilon$, where ϵ is a small value, will get close as $t \rightarrow \infty$.

2. A **source** is an unstable equilibrium, that repel neighboring solutions.
3. A **node** is an unstable equilibrium, which is all the other cases.

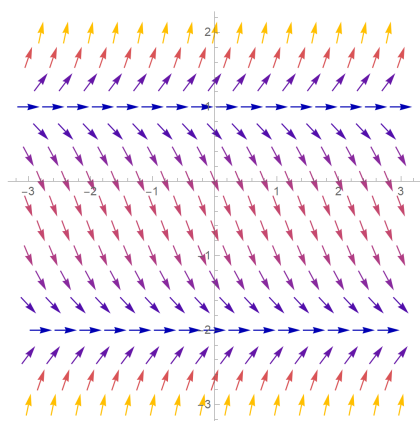
Definition 2.5. Since an autonomous differential equation doesn't depend on the independent variable, its Direction Field looks the same for any value of t (or x). Therefore, the same information can be represented in a compact form - on a y axis only, which we call a **Phase Line**.

The equilibrium points are represented as circles on a phase line. The increasing slope is represented as an arrow up, while the decreasing as an arrow down. As such, a sink will be surrounded by arrows pointing inside a circle, while sources are surrounded by arrow pointing out.

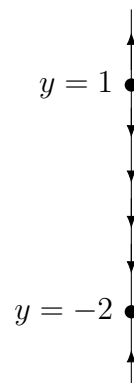
Example 2.6. Consider

$$y' = (y - 1)(y + 2).$$

The vector field has two equilibrium lines, the zeros of the slope $y = 1$ and $y = -2$. When $y > 1$ the slope is clearly positive, when $y < 1$ the slope is negative. When



(a) Direction Field



(b) Phase Line

Figure 3: Direction Field and a Phase Line of $y' = (y - 1)(y + 2)$

$y < -2$ both $y - 1 < 0$ and $y + 2 < 0$, hence the slope is positive, see Figure 3a. From the Figure 3a one can notice that $y = 1$ is a source and $y = -2$ is a sink. An equivalent Phase Line representation is given in Figure 3b.